

Demagnetization and Thermal Analysis of Six-Phase Permanent Magnet Motor-based Integrated Onboard EV Charging

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Abstract—Integrated onboard battery charger (IOBC) is an emerging technology that charges electric car battery packs utilizing the same circuitry for electric propulsion, reducing the cost and weight of an EV. During charging, while ensuring zero-average torque to avoid rotation, permanent magnets (PMs) synchronous motors with fractional slot concentrated winding (FSCW) may be subjected to irreversible demagnetization due to sub and super space harmonics with relatively high magnitudes. This paper presents the design and optimization of six-phase surface-mounted permanent magnet (SPM) and interior permanent magnet (IPM) motors with FSCW considering the impact of PM demagnetization during IOBC. A detailed thermal and demagnetization analysis is conducted to identify the risk of PM demagnetization during propulsion and IOBC modes. The performance and demagnetization risk for SPM and IPM motors with different power ratings used in commercial EVs are compared at various operating conditions. In the integrated charging process, grid-to-vehicle and vehicle-to-grid power circulations are practically implemented. Simulations are conducted using a three-dimensional finite element analysis. Experimental results of a 15 kW IPM motor are provided for further verification.

Index Terms—Fractional slot concentrated winding, PM Demagnetization, Integrated onboard charger, six-phase PMSM, thermal analysis.

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I. INTRODUCTION

ELECTRIC vehicles (EVs) are proving to be an excellent investment in the form of smart, reliable, and environmentally friendly electrical infrastructure. Bidirectional chargers enable EVs to function as mobile energy storage assets which can supply power back to homes, buildings, and the grid through vehicle-to-home/building (V2H/B), grid-to-vehicle (G2V), and vehicle-to-grid (V2G) technology [1]. Combining this technology with an integrated onboard battery charger (IOBC) for EVs is of paramount importance since the IOBC eliminates the need for a charging infrastructure by utilizing the onboard powertrain elements to charge the batteries. Consequently, a significant obstacle to EV adoption will be reduced [2]. In IOBC technology, the windings of the traction motor serve as line inductors and the accompanying drive circuitry as AC/DC converters which reduce the weight and cost of the EV by eliminating the need for a dedicated charger. Moreover, relatively fast charging can be achieved with the help of various proposed charging topologies [3].

IOBCs can be classified into isolated and non-isolated configurations. Non-isolated IOBC tend to have smaller size and lower weight compared to the isolated counterparts. However, the chargers having galvanic isolation are more attractive in terms of electrical safety. To overcome isolation safety problems, various possibilities have been investigated with an emphasis on an electric machine configuration with an extra set of windings. A non-isolated single-phase integrated battery charger was proposed in [4]. It has been installed on an electric scooter prototype and uses the propulsion inverter with an additional power rectifier and an LC filter. The most preferable option to solve the galvanic isolation issue for the presented and similar topologies is installing isolated line frequency transformers in the charging points, which will be off-board and has nothing to do with the onboard components [3]. However, most of the integrated configurations proposed so far are non-isolated [5]. Thus, for non-isolated chargers, it is essential to properly manage the common-mode current. Despite of isolation issue, non-isolated integrated charging will remain a very attractive option for future electric vehicles because of the potential to achieve charging, V2G, and V2H functions at a lower cost, weight, and volume [6].

Any inverter-fed motor such as induction motors, permanent magnet synchronous motors (PMSMs), and switched reluctance motors (SRMs) can be used for the IOBC application. PMSMs are the most attractive candidates for traction in EVs due to their high torque density, high power density, wide speed range, and relatively precise control [7]. Surface-mounted PM (SPM) and Interior PM (IPM) machines have been successfully applied in various commercially used EVs [8]. In order to prevent mechanical rotation during charging mode, the motor's winding should be energized such that the average torque produced by the motor is zero. This can be accomplished by either applying an adequate control system or changing the winding arrangement [9]. Three-phase motors in general exhibit non-zero torque in IOBC due to the rotating magnetic field produced by the three-phase AC currents, thus, requiring extra/redundant hardware setup to avoid mechanical rotation during charging mode [10].

Multiphase motors have many advantages over three-phase drives, including lower per-phase current rating; stator excitation in a multiphase machine produces an improved magnetomotive force (MMF), resulting in lower space harmonics, lower torque ripples, and higher efficiency than three-phase motor drives [11]. Such excitation generates pulsating torques at even multiples of the fundamental excitation frequency. Furthermore, employing a multiphase motor minimizes dc-link harmonics, reduces the required dc-link capacitance, and enhances fault-tolerant operation since the degrees of freedom increase as the number of independent phases increases [12]. Using multi-phase motors, zero-average torque production can be achieved during charging with minimum hardware change. For instance, five-phase, six-phase, and nine-phase motors are used in the literature to implement IOBC applications [13]. Six-phase motor seems to be the most attractive candidate for IOBC where the torque can be cancelled by managing two sets of dq -axis currents [9]. Although the average mechanical torque generated by IOBC operation can be eliminated by managing winding configuration and control to cancel out the torque-producing flux component, the harmonic flux components produced continue to exist in many circumstances, which represent the main source of torque ripple and core losses and therefore, should be managed carefully for minimizing losses, vibration, and ensure optimal charging.

The slot/pole combination selection is also crucial in EV applications. In [14], several viable six- and nine-phase slot/pole combinations are presented. The characteristics and performance of different slot/pole combinations and various IOBC topologies are summarized. The fractional slot concentrated winding (FSCW) 12-slots/10-pole combination exhibits an improved winding factor and lower rotor losses in both propulsion and charging modes. Moreover, FSCW has shown promising results in EV applications owing to their advantages. These advantages include a high slot fill factor, short-end turns, flux-weakening, and fault-tolerance capabilities [15], [16]. The FSCW-based six-phase PMSM shows satisfactory performance by achieving zero-average torque in the charging mode of operation [9]. However, the sub and super-synchronous harmonic flux components can be still found in the air gap that cannot be canceled out completely like in distributed winding configuration. These harmonics exerted on the rotor have an adverse influence on the PMs of the PMSM. Thus, along with zero-average torque, vibration, current harmonic reduction, and unity power factor, the magnetic flux

density of the PM should also be carefully monitored. Furthermore, the thermal impact on the winding and PMs must be carefully considered due to poor ventilation during charging mode. All these factors contribute to overheating yielding lower efficiency and inevitable PM irreversible demagnetization [17].

In recent years, considerable work has been done highlighting the thermal modeling and analysis of PMSMs and induction motors, particularly in the propulsion mode [18]–[21]. In [18], a comparative study on thermal modeling and design of PMSMs and IMs is presented. Accordingly, the effect of several thermal parameters on the temperature of the windings, PMs, and rotor bars has been assessed based on thermal network models and a comprehensive sensitivity study. As a result, the effect of the heat transfer coefficient of the primary cooling is more crucial in IM than their PM counterpart. Moreover, thermal analysis of high-speed IPM machines is conducted based on the computational fluid dynamics in [19]. Thermal analysis of a six-phase PMSM with asymmetric winding is presented in [22]. However, this study only considered the propulsion mode. Similarly, a detailed demagnetization study of SPM and IPM motors is conducted in [23]. However, this study only considers the propulsion mode. Furthermore, another demagnetization analysis for a three-phase IPM-type machine is conducted in [24]. However, three-phase motors are not the most attractive candidates for IOBC, and this study does not give details about the thermal analysis and the implementation of the charging mode. Most of the existing studies on thermal modeling and demagnetization are carried out for motors running in motoring mode. However, when the machine is used as a filter in the IOBC application, the effect of the winding current on the generated heat and temperature rise and the effect of the leakage flux components due to harmonic flux components on the magnets have not been studied thus far. Therefore, a comprehensive thermal and demagnetization analysis of both SPM and IPM-type machines during IOBC operation is necessary to evaluate the risk of demagnetization. The contributions of this paper are summarized as follows:

- Designing six-phase FSCW SPM and IPM-type motors of different power ratings for propulsion and IOBC application for EVs.
- Modifying the multi-objective genetic algorithm (MOGA) to optimize the motors considering the impact of the demagnetization in IOBC mode.
- Conducting a detailed demagnetization and thermal analysis at different operating conditions and analyzing the impact of demagnetization due to IOBC in SPM and IPM motors.
- Comparing the risk of demagnetization for SPM and IPM motors at different power ratings and charging levels.
- Experimental verification using a 15 kW IPM prototype machine.

The remainder of the paper is structured as follows. The theory and functioning of the IOBC are illustrated in Section II, and the design and optimization of the motors are covered in Section III. Section IV presents the thorough thermal and demagnetization study as well as the calculated risk of demagnetization. Finally, Section V presents the experimental findings that validate the analysis, and Section VI concludes the discussion.

II. IOBC USING SIX-PHASE FSCW PMSM

An IOBC can manage a bidirectional power flow without an extra onboard charger unit. It uses the same power electronics circuitry that is used for propulsion to draw power from the grid to charge the battery in G2V mode and the opposite in V2G mode. Furthermore, it can also be used in V2H/B mode to provide power to a house or building in the event of a blackout or some other disaster [25]. The implementation of V2G mode is more complex due to several constraints of the main grid. However, the V2H or V2B and other similar technologies can be easily implemented with improved cost and less complexity by utilizing IOBC.

A six-phase PMSM with asymmetrical winding is considered the best candidate for EV applications due to higher current

capability that can be helpful in faster charging and high torque in propulsion, better fault-tolerant capability, zero-average torque during charging, and require less hardware reconfiguration to switch between propulsion and charging modes [6]. Fig. 1(a) illustrates the schematic representation of the IOBC proposed in this analysis for propulsion and charging. It consists of a six-phase machine, a six-phase inverter, and a battery. The primary goal in charging is to achieve a zero-average torque to avoid mechanical rotation. Moreover, a low peak-to-peak torque ripple is required to reduce vibrations and noise. Therefore, the windings need to be energized in such a way that the magnetic flux produced is cancelled out to prevent mechanical rotation and maximize the grid line current. The phasor diagrams of the winding spatial orientation in propulsion and charging modes are shown in Fig. 1(b) and (c), respectively. The two three-phase winding sets in an asymmetrical six-phase winding topology are separated by 30° in propulsion and 150° in charging mode [6]. To implement charging with minimum winding rearrangement while ensuring zero-average torque, the end terminal of $(a_1$ and $b_2)$, $(b_1$ and $c_2)$, and $(c_1$ and $a_2)$ are connected as shown in Fig. 1(a). In this case, the windings are adjusted such that the flux produced by one three-phase winding group cancels out the effects of the other group, guaranteeing zero-average torque. The transition between propulsion and charging mode is achieved by switches $(S_1$ – S_5). The charging process begins when S_1 is switched on and S_2 – S_5 are turned off, and the opposite is the case for the propulsion mode. In charging mode, the grid line currents i_a^g , i_b^g , and i_c^g are divided between the first three-phase winding set $(i_{a1}$, i_{b1} , and $i_{c1})$ and the second three-phase winding set $(i_{a2}$, i_{b2} , and $i_{c2})$, respectively, following synchronization. As a result, the resulting magnetic fields from the two winding groups oppose each other, generating a zero torque-producing magnetizing flux component. The six-phase windings are utilized as grid-side filters. IOBC not only improves the design, weight, and cost of the EV by eliminating the conventional onboard charger but also improves the charging time compared to level 1 and level 2 charging by due to the capability of high power.

III. OVERVIEW OF MOTOR DESIGN OPTIMIZATION

Six-phase SPM and IPM motors rated at 15 kW are specially designed to test the propulsion and charging modes. The requirement, input parameters, and constraints for both target motors are given in Table I. Asymmetrical winding arrangement are employed due to its superior performance over the dual three-phase. The initial designs were achieved by using a parametric magnetic equivalent circuit (MEC) model [26]. First, the sizing equations are used to determine the initial machine dimensions. Then, the number of turns and the rated RMS current are defined based on the rated power and DC link voltage. Finally, the electromagnetic performances can be obtained from the MEC model. To achieve the optimal design that is suitable for propulsion and charging in an EV, both SPM and IPM models are optimized. A multi-objective optimization problem is formulated to define the optimal machine geometrical dimensions based on the efficient multi-objective genetic algorithm (MOGA) [27]. Initially,

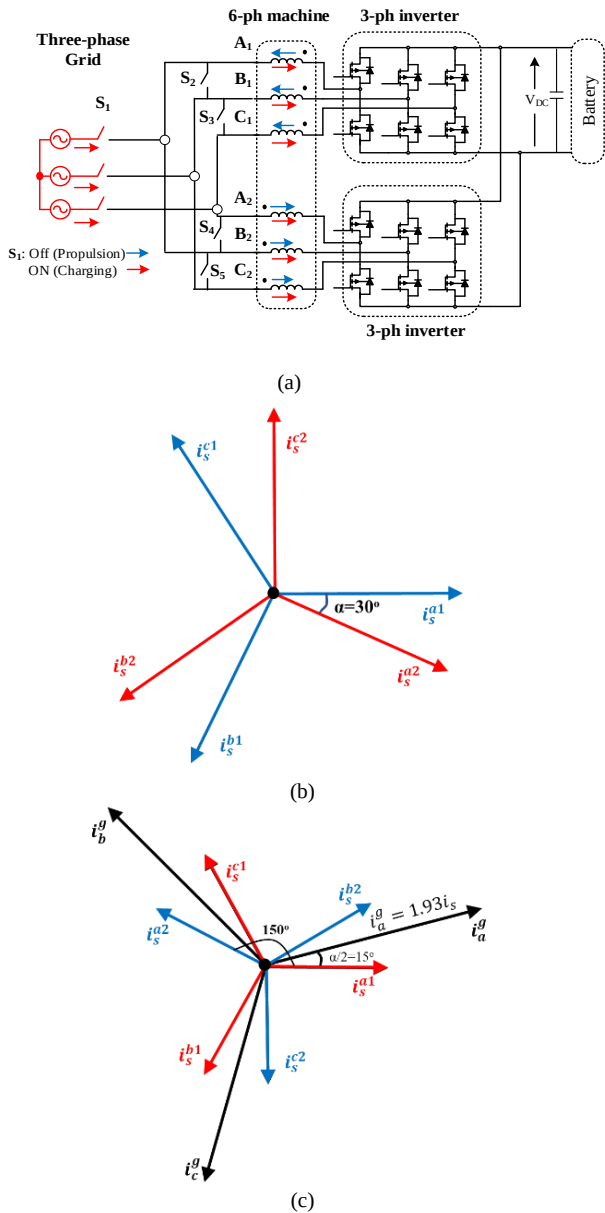


Fig. 1. Six-phase IOBC schematic and phasor diagram. (a) Six-phase asymmetrical configuration, (b) Phasor diagram under propulsion, (c) Phasor diagram under charging mode.

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the optimization objectives, decision variables, and constraints are determined.

The main criteria to allocate optimal design – besides average torque production (T_{mean}^{prop}), torque ripple (T_{ripple}^{prop}), and core losses (P_{core}^{prop}) in the propulsion – are the maximum magnetic field

TABLE I
REQUIREMENTS AND CONSTRAINTS OF THE TARGET MOTOR DESIGN

Parameter	Unit	Values
SPM/IPM		
Case-study EV requirements		
Rated power	kW	15
Rated speed	rpm	1500
Maximum speed	rpm	4000/7500
Rated torque	Nm	95.5
Phase current	A	18
Inputs		
Number of Phases		6
Number of slots		12
Number of poles		10
Airgap flux density	T	0.8
Stator electrical loading	A/mm	12
Constraints		
Required flux density	T	1.5
Current density	A/mm ²	4
Copper filling factor		0.45
DC bus voltage	V	600
Min. efficiency at rated speed	%	95

TABLE II
FINAL DESIGN PARAMETERS OF THE SPM AND IPM MOTORS

Parameter	Unit	Values
SPM / IPM		
Outer Diameter	mm	279.4
Stack Length	mm	147.7
Length of Air Gap	mm	1/0.5
Magnet Type	-	N40H
Rotor Core Material	-	M250-35A
Machine Volume	mm ³	9.06×10 ⁶
Number of turns	-	27

intensity (H_{max}^{charg}), torque ripple (T_{ripple}^{charg}), and core losses (P_{core}^{charg}) under charging mode. The contribution of this study in design optimization is to include the maximum magnetic field intensity in charging mode (H_{max}^{charg}) in the objective function and compare it with the coercivity (H_c) to minimize the risk of PM demagnetization. Several geometrical dimensions are varied to define the optimal design, such as the magnet thickness (Y_m), tooth-tang depth (d_1), core back width (Y_{sb}), tooth width (W_t), slot opening ratio (t_{so}/τ_{so}), and PM width to pole pitch ratio (α_{PM}). The optimization problem is formulated as follows:

$$\min_{X_i} F(X_i) = \lambda_1 \frac{T_{mean}^{prop}(X_i)}{T_{mean}^{prop}(X_i)} + \lambda_2 \frac{T_{ripple}^{prop}(X_i)}{T_{ripple}^{prop}(X_i)} + \lambda_3 \frac{P_{core}^{prop}(X_i)}{P_{core}^{prop}(X_i)} + \lambda_4 \frac{H_{max}^{charg}(X_i)}{H_{max}^{charg}(X_i)} + \lambda_5 \frac{T_{ripple}^{charg}(X_i)}{T_{ripple}^{charg}(X_i)} + \lambda_6 \frac{P_{core}^{charg}(X_i)}{P_{core}^{charg}(X_i)} \quad (1)$$

where $\vec{X}_i = \left[Y_m, d_1, Y_{sb}, W_t, \frac{t_{so}}{\tau_{so}}, \alpha_{PM} \right]$

$$\begin{aligned} & J \leq 4 \text{ A/mm}^2 \\ & B_{tooth} \leq 1.7 \text{ T} \\ & T_{mean} \geq 95.5 \text{ Nm} \\ & T_{ripple} \leq 8 \% \\ & X_i^{min} \leq X_i \leq X_i^{max} \end{aligned} \quad (2)$$

Where B_{tooth} is the flux density in the tooth tips. The initial values of the above-mentioned optimization objectives are T_{ripple}^{prop} , T_{core}^{prop} , f'_{cost} , H_{max}^{charg} , T_{ripple}^{charg} , and P_{core}^{charg} , respectively. Furthermore, the corresponding weighting factors are $\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5$, and λ_6 , respectively. It is worth mentioning that $\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5 + \lambda_6 = 1$. In this paper, the six objectives are of the same importance. Thus, the same weight factor is utilized.

Thereafter, the influence of each design variable on the optimization objectives is studied based on the comprehensive sensitivity analysis [28]. Accordingly, the high-sensitive design parameters are defined, and their variation range is enhanced through the Response Surface Methodology (RSM). Eventually, the MOGA defines the optimal trade-off amongst the proposed objectives. The optimal design point is therefore allocated. The overall design optimization process has not been elaborated to avoid repetition and can be revisited in [29]. For example, the optimal design is specified for the SPM machine under both operational modes, as shown in Fig. 2.

The optimized design parameters are revealed in Table II. For a fair comparison, the same stack length, stator outer diameter, and copper fill factor are utilized for the proposed machines. The 3D finite element models (FEM) of SPM and IPM motors are shown in Fig. 3. The electromagnetic torque generated by both motors under propulsion and charging modes is compared in Fig 4(a) and (b), respectively. As anticipated, the IPM motor produced higher average torque (106.7 Nm) in propulsion mode than the SPM motor (96.4 Nm). The SPM motor, however, has less torque ripples. In charging mode, zero-average torque was achieved by both machines, however, like in propulsion, the peak-to-peak torque of IPM (5.6 Nm) was higher than that of SPM (2.3 Nm).

The loss model in Ansys Maxwell is used to determine the losses of the two motors [30]. Core and PM losses during charging mode are compared in Fig. 5. Core, PM, and copper losses contribute directly to the temperature rise of the motor. The copper and stator core loss makes up most of the motor's total loss. According to theoretical analysis, the stator current has a significant impact on the air gap magnetic flux density distribution of the IPM motor due to its small air gap length. As a result, its harmonic components of the air gap flux density are much larger than those of the SPM motor, leading to a larger stator core loss. Further performance detail of both machines is summarized in Table III. The main reason that the core losses are much lower in charging than that during propulsion is mainly due to the operation frequency, i.e., the frequency during propulsion (125 Hz) depends on the speed and number of pole pairs, while the frequency during charging is equal to the grid frequency of 50 Hz. Since the core losses depends on the frequency, the core losses during charging are much lower than that during propulsion.

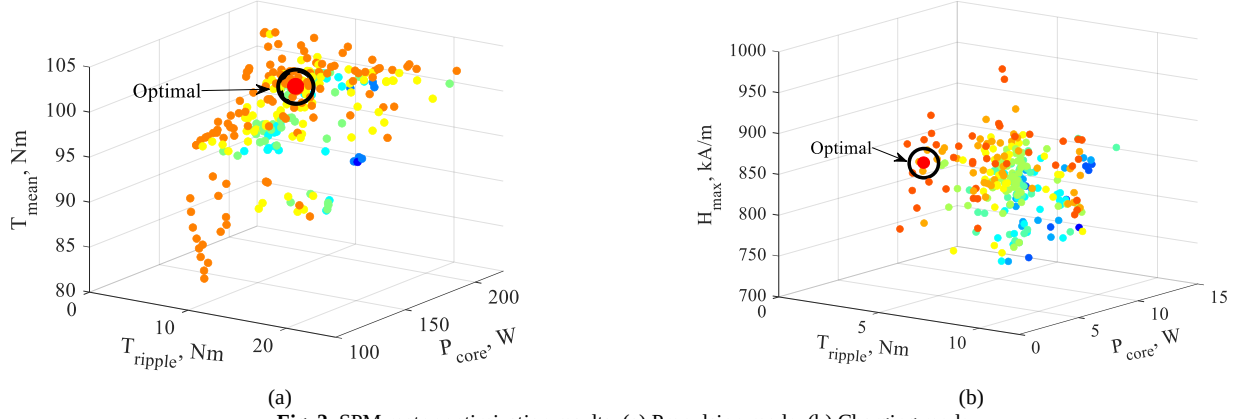


Fig. 2. SPM motor optimization results. (a) Propulsion mode. (b) Charging mode.

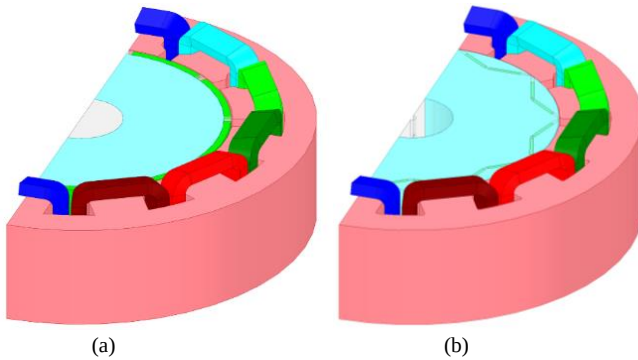


Fig. 3. 3D FEM machine models. (a) SPM. (b) IPM.

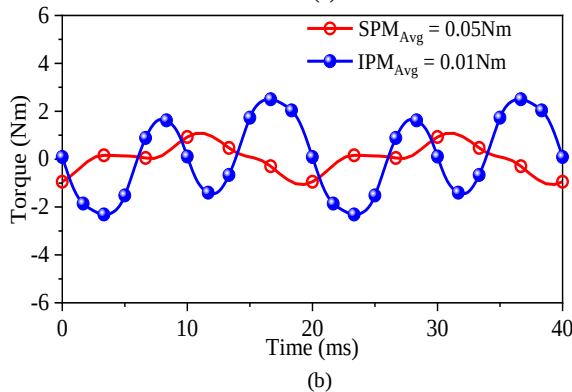
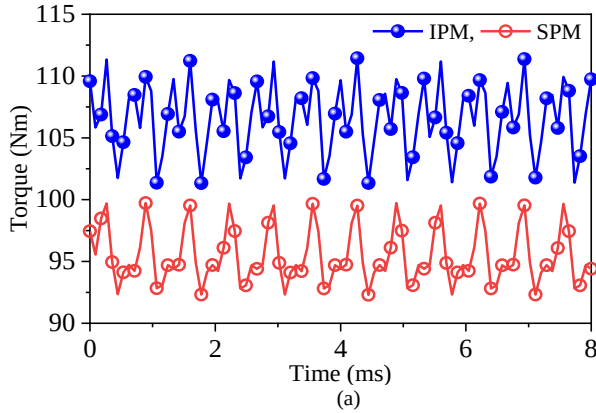


Fig. 4. 3D Electromagnetic torque generated for SPM and IPM machines. (a) during propulsion mode (b) during charging mode.

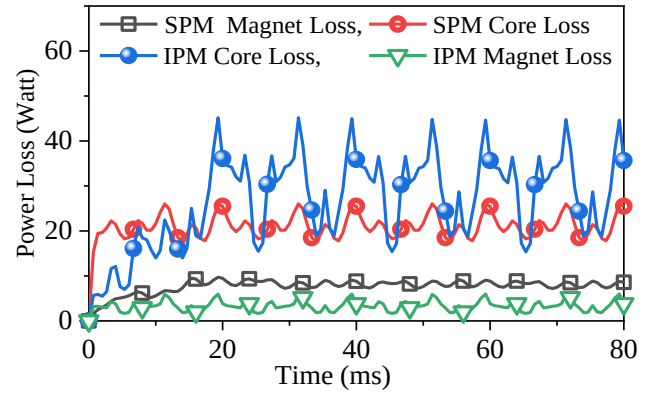


Fig. 5. Core and PM losses comparison in SPM and IPM motor during IOBC mode.

TABLE III
PERFORMANCE COMPARISON OF SPM AND IPM MOTORS

	Propulsion (base RPM)		Charging	
	SPM	IPM	SPM	IPM
Average Torque (Nm)	96.4	106.7	0.05	0.1
Pk-pk Torque (Nm)	7.7	9.3	2.3	5.6
Current (A)	18	18	18	18
Phase Voltage	306	310	15	32
Core loss	182	254	25	35
PM loss	304	47	24.5	5.9

IV. THERMAL AND DEMAGNETIZATION ANALYSIS

In the previous section, the motors were optimized considering the PM demagnetization effect during the integrated charging mode. In this section, a coupled electromagnetic and thermal analysis is presented. Demagnetization shortens the lifetime of a motor since it reduces the machine's performance and dependability over time. Demagnetization can occur due to various conditions, including temperature, overload, mechanical stress, and aging [31]. Consequently, it is quite difficult to consider all these elements in a single model. The desired method for analyzing the demagnetization behavior while taking into account all pertinent elements is a multi-physics approach that couples the motor's electromagnetic and thermal performances and examines how they interact. Fig. 6 illustrates the process of the entire study to analyze thermal and demagnetization risk. The first step is obtaining an optimized design and calculating the

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losses which are discussed in the previous sections. The next step is to perform a detailed thermal and demagnetization analysis which will be discussed in this section. Finally, the experimental verification will be presented in the following section.

A. Thermal Analysis

Keeping the electric motor at the proper temperature is another critical component. When an electric motor overheats, its energy consumption increases significantly. In PM machines, overheating causes thermal demagnetization. As a result, electric traction motors' heat management is of paramount importance especially the PM temperature which cannot be measured directly. A detailed thermal analysis of both the SPM and IPM machines was conducted to accurately estimate the temperature of the PMs. Ansys Mechanical simulation platform is used to calculate the temperature distribution of both machines in charging mode. Since the rotor remains stationary during charging, there is no thermal effect due to rotation. Therefore, Ansys Mechanical is a suitable platform for thermal analysis. Since the target application in this study is IOBC of a small EV, the benchmark machines fall in the total enclosed nonventilated (TENV) category where the heat is dissipated by the external surface of the housing by natural convection. An accurate heat convection coefficient is required to compute temperature distribution in a motor. The convection coefficient depends upon the geometry of the surface as well as its orientation. It also depends on the variation in temperature on the surface and the thermophysical properties of the fluid involved. The geometry of the motor is close to a cylinder in a horizontal orientation in this study. The fluid in natural convection involved is air. Thus, the convection coefficient can be calculated using the detail given in [32] and [33].

Iron, copper, and eddy current losses are the main sources of heat in a motor. Thermal simulations were performed when the motor was operating under charging mode. In all simulations, reference temperatures of 20° and 40°C are used as reference temperatures. The contour plot of temperature distribution in different parts of the SPM motor model is given in Fig. 7. Table IV shows the temperature of each component of the SPM machine at various ambient temperatures in charging and propulsion modes. The winding temperature was measured to be 56.2°C and 102°C in the charging and propulsion modes, respectively. While the stator yoke reached 36 °C, the rotor lamination was 34.3 °C, and the PM 34.4 °C during charging. Similarly, Fig. 8 depicts the temperature distribution in the IPM motor operating in IOBC mode. The maximum temperature in the windings was 58.2 °C, whereas in the PMs 31 °C was recorded during charging. Table V shows the temperature distribution of each component at various reference temperatures in detail. The simulation findings reveal that the PM temperature during integrated IOBC mode can easily approach 60 °C. However, in propulsion mode, both machines had higher temperatures than in IOBC mode. Eddy current loss contributes significantly to heat generation in the rotor area. Furthermore, the heat is a function of the motor's speed in propulsion mode. The winding and PM temperatures on the SPM machine reached 122 °C and 85 °C, respectively.

Similarly, the IPM machine followed the same pattern, with winding and PM temperatures reaching 126 °C and 84 °C, respectively. The reason for the lower temperature in the IPM machine is due to lower permanent magnet loss compared to the SPM machine.

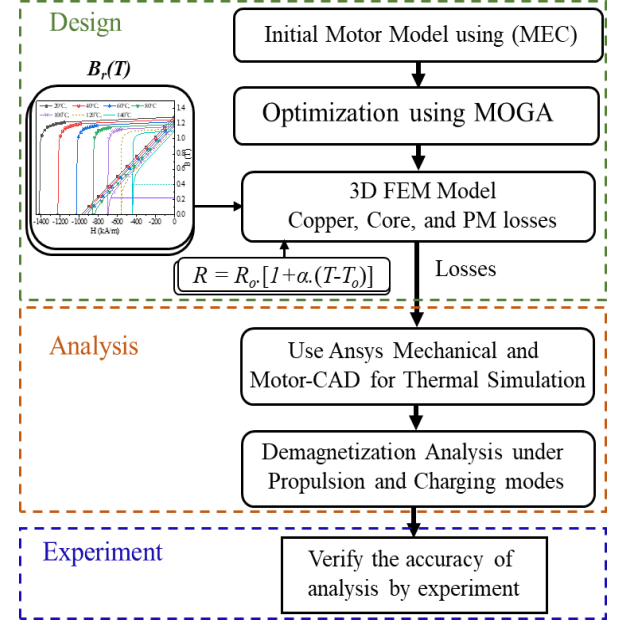


Fig. 6. Flow chart of the entire analysis.

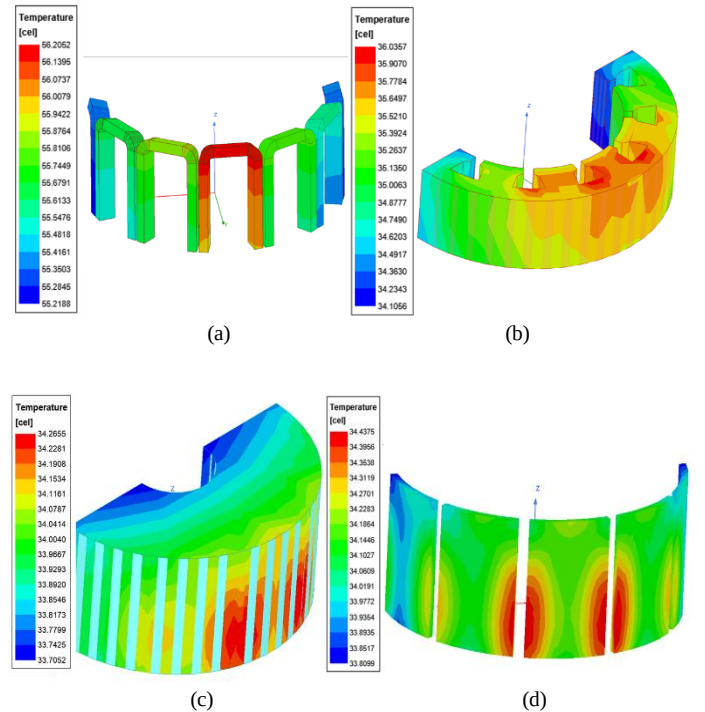


Fig. 7. Temperature distribution in SPM type machine during charging mode. (a) Winding, (b) stator, (c) rotor, (d) permanent magnets.

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TABLE IV
TEMPERATURE DISTRIBUTION OF SPM MOTOR UNDER PROPULSION AND CHARGING

Integrated Onboard Charging Mode				
Ref. Temperature	Stator yoke [°C]	Temp. Winding [°C]	Rotor lamination [°C]	Magnet [°C]
20°C	36	56.2	36	34.4
40°C	57.4	82.9	54.3	54.8
Propulsion Mode				
20°C	73	102	62.6	65.4
40°C	98	122.7	82.7	85.5

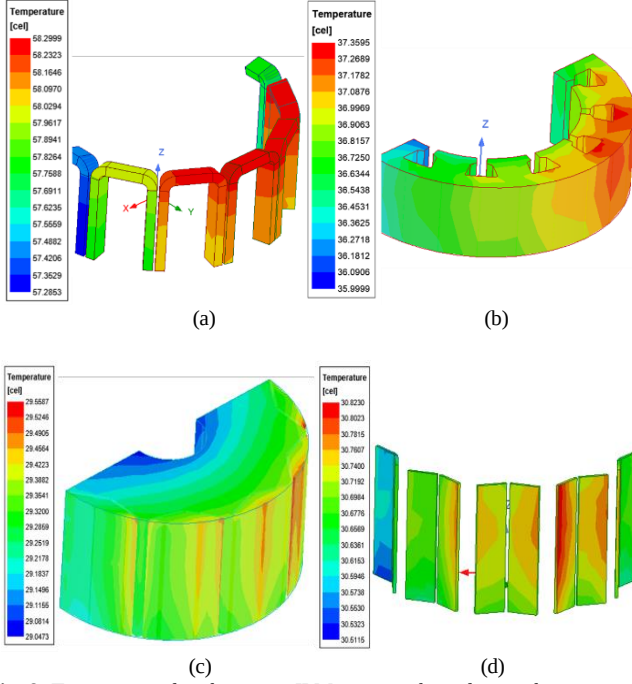


Fig. 8. Temperature distribution in IPM-type machine during charging mode. (a) Winding, (b) stator, (c) rotor, (d) permanent magnets.

TABLE V
TEMPERATURE DISTRIBUTION OF IPM MOTOR UNDER PROPULSION AND CHARGING

Ref. Temperature	Stator yoke [°C]	Winding [°C]	Rotor lamination [°C]	Magnet [°C]
IOBC-Mode				
20°C	31	62	37.3	31
40°C	51.9	82.2	48.3	41.2
Propulsion Mode				
20°C	77	106.6	67.6	64.4
40°C	97.3	126	87.8	84.7

B. Demagnetization Analysis

The irreversible demagnetization of PMs is regarded as one of the most significant constraints in PM-type machines. Excessive demagnetizing magnetomotive force (MMF) during zero mechanical torque acquisition for IOBC can have a negative impact on the PMs. During IOBC operation, the d -axis of the rotor and the magnetic axis of the stator may overlap due to control mechanisms for minimizing undesired torque. The risk of irreversible demagnetization is much higher under such conditions compared to propulsion mode.

In steady-state operation, the rise in temperature due to overloading, improper control, and/or any type of fault in the

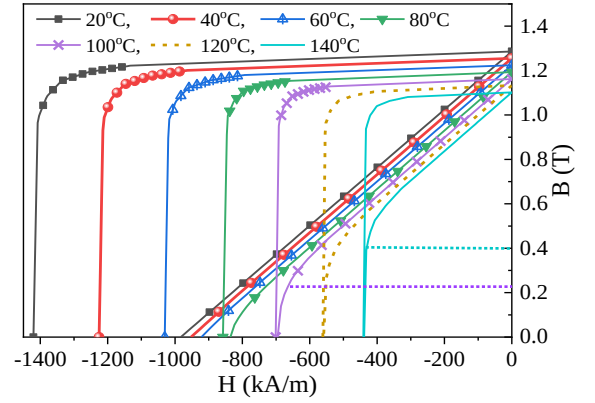


Fig. 9. Intrinsic BH curve of the N40H permanent magnet.

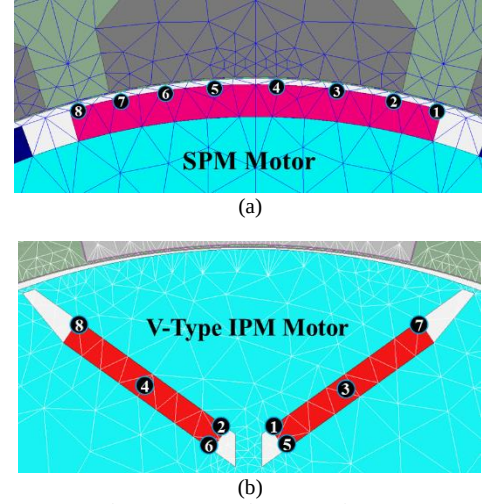


Fig. 10. Points at PMs for calculation of magnetic flux density in SPM and IPM machines.

motor can lead to irreversible demagnetization. Fig. 9 depicts the intrinsic and normal characteristic curves of the Nd-Fe-B (H grade) PM at various temperatures. As the temperature rises, the residual magnetic flux density (B_r) drops, and the demagnetization cutoff point or knee point rises. This illustrates that at higher temperatures, the operating point region of a BH curve narrows, and as a result, the probability of demagnetization increases significantly. The thermal results of the benchmark machines reveal that the temperature can reach up to 130 °C at rated speed and load. The temperature of the PMs reaches up to 85 °C at rated speed. However, at maximum speed, the core and PM losses increase significantly. Consequently, the PM temperature can easily reach up to 130 °C. Furthermore, the literature indicates that the working temperature of any EV's motor can be raised to around 140 °C [10]. To investigate the worst-case scenario, the maximal temperature in this study is set at 140 °C [34].

To precisely investigate demagnetization, FEA is used to identify the weakest locations on the magnets, and then eight separate points on the PMs of both machines are selected for monitoring, as shown in Fig. 10. The magnetic field of the stator has a direct influence on the top area of an SPM machine. As a result, the entire area of the PMs towards the airgap is deemed sensitive, hence the top portion of the magnet was designated for monitoring, as shown in Fig. 10(a). In IPM motors, the upper and lower edges close to the flux barrier are normally vulnerable. However, the trend may be different in charging mode since the

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flux distribution is not the same. Thereby, all the corners and the center of the PMs in the IPM machine were designated for monitoring, as shown in Fig. 10(b). Since both benchmark machines are radial-flux ones, only the radial component of the magnetic field intensity (H_{radial}) and magnetic flux density (B_{radial}) given in (3) and (4) will contribute to demagnetization. Where H_{xy} and B_{xy} are the x- and y-axis components of the magnetic field intensity and PM flux density, respectively. θ is the angle between the magnetization angle and the x-axis of phase-A.

$$H_{radial} = H_x \cos \theta + H_y \sin \theta \quad (3)$$

$$B_{radial} = B_x \cos \theta + B_y \sin \theta \quad (4)$$

The magnetic flux density distribution in the PMs of SPM and IPM motors under IOBC mode is obtained using finite element analysis. In Fig. 11, the minimal magnetic flux density (B_{min}) of the SPM machine at a rated current and temperature of 100 °C is highlighted. The B_{min} is around 0.4 T in relation to the knee-point of 0.215 T. The demagnetization curve for the SPM machine in propulsion and IOBC mode is determined and shown in Fig. 12. At 100 °C and 140 °C, the radial component of the PM flux density is obtained at the weakest point. In propulsion mode, the PMs of the SPM machine are safe with $B_{min} = 0.55T$ versus the knee point (0.28 T) at 100 °C, however, in the worst-case scenario (140 °C), the B_{min} is 0.51T, which is less than the knee-point (0.56T). As a result, irreversible demagnetization has occurred, and the magnetic flux density of the PM will not recover to its original value (B_r) but will instead follow the T_2 trajectory and recover depending on recoil permeability. Similarly, the variation of the operating point in charging mode is depicted in Fig. 12. (b).

At 100 °C, the PMs are in a safe region; nevertheless, the B_{min} is lower than in propulsion mode, indicating that the PMs are under higher stress. Furthermore, at 140 °C, the magnetic flux density fell dramatically to 0.43 T, much below the knee point. As a result of the extreme irreversible demagnetization, the residual magnetic flux density will never return to its initial value.

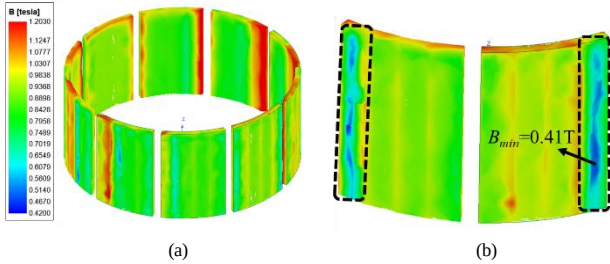


Fig. 11. Magnetic flux density distribution in SPM motor under IOBC. (a) All PMs, (b) PMs with B_{min} .

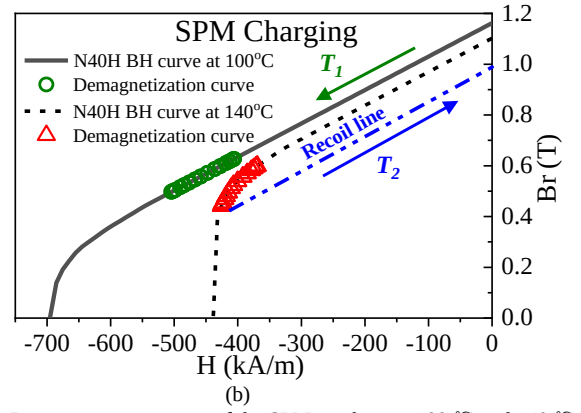
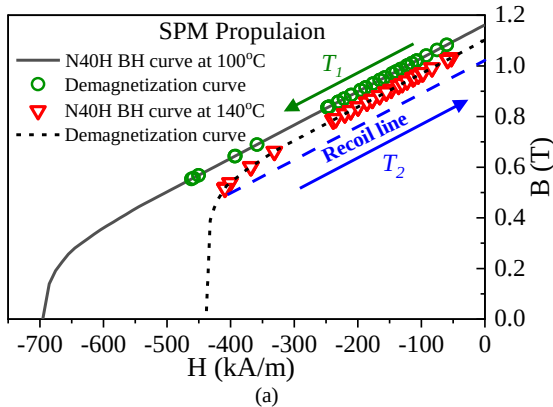


Fig. 12. Demagnetization curve of the SPM machine at 100 °C and 140 °C. (a) Propulsion mode, (b) Charging mode.

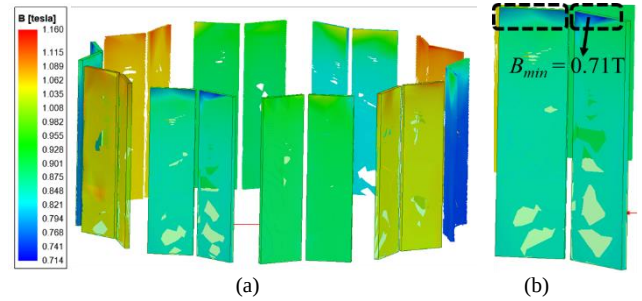


Fig. 13. Magnetic flux density distribution in IPM motor in IOBC. (a) All PMs, (b) PMs with B_{min} .

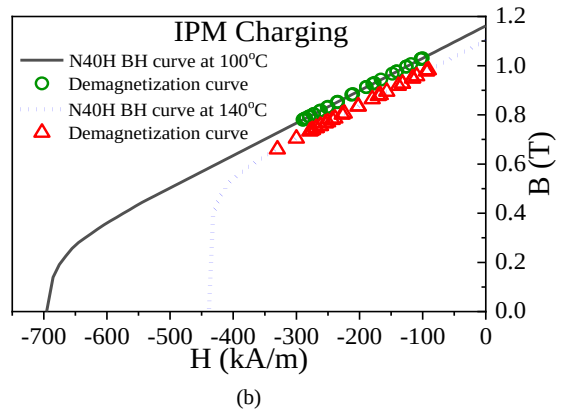
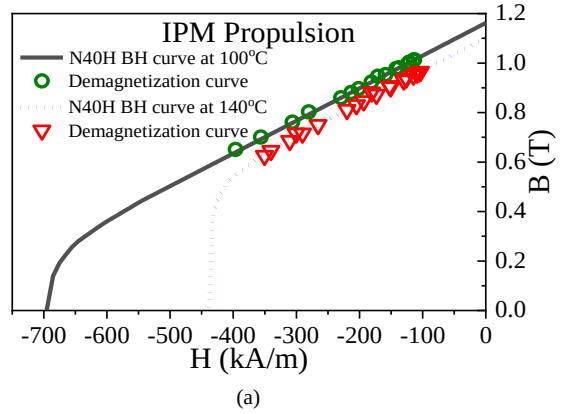


Fig. 14. Demagnetization curve of the IPM machine at 100 °C and 140 °C. (a) Propulsion mode, (b) Charging mode.

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The B_{min} of the IPM motor at a rated current and temperature of 100 °C is highlighted in Fig. 13, indicating that the IPM machine has a larger buffer before irreversible demagnetization than the SPM one. Fig. 14 shows the demagnetization curve of the IPM machine in propulsion and charging modes. The IPM machine appears to be substantially safer than SPM. Even at 140 °C, the machine is operating above the knee. The B_{min} is greater than 0.6 T, as shown in Fig. 14, which is substantially greater than the knee point. This demonstrates that the stress on the buried PMs in the IPM machine is rather modest. Although, the PMs of the IPM machine are safer in the worst case. However, the overall efficiency of the motor in such high temperatures will be severely reduced. Therefore, a proper cooling mechanism is normally used in EVs.

As previously stated, overloading or inrush current can also potentially cause damage to PMs. To avoid damage from a transient, each PM-type motor should be able to resist at least three times its rated current. Fig. 15(a) and (b) demonstrate the variation in B_{min} at the investigated points for various stator currents during charging for SPM and IPM motors, respectively. There is a considerable reduction in flux density at edges (points 1, 2, 7, and 8) of the SPM machine even at rated current. As a result, any transient or increase in temperature above the benchmark will quickly demagnetize the magnets.

On the other hand, the strength of the IPM machine's magnets at all selected points demonstrated robust performance against the demagnetizing MMF. Even at 80 A stator current, the flux density

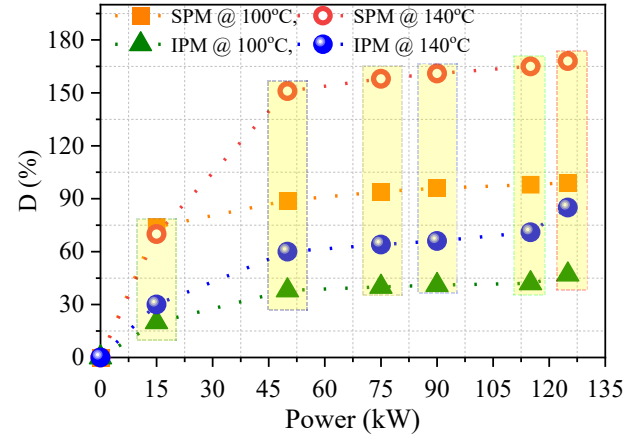


Fig. 16. Demagnetization risk comparison in motors of different power ratings at different temperatures under charging.

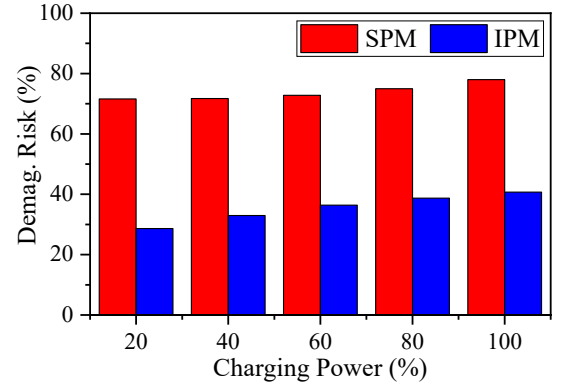
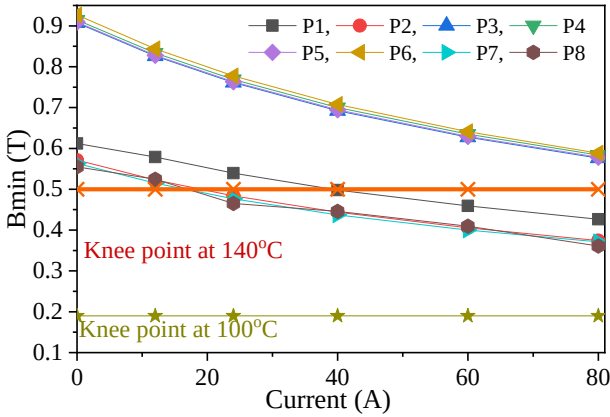


Fig. 17. Demagnetization risk assessment in 15 kW SPM and IPM motors with charging levels.

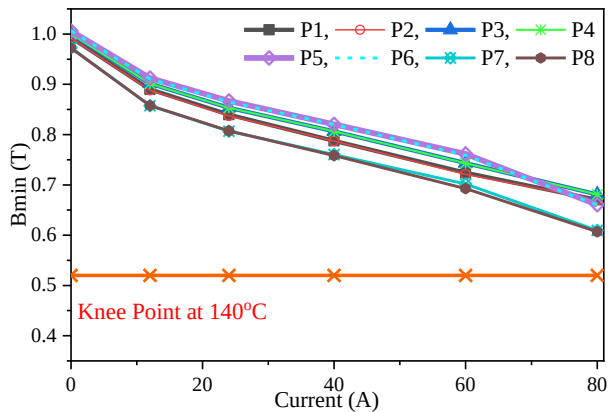
of the PMs is above the knee-point, indicating that there will be no risk of demagnetization in this machine in both steady-state and transient, but the risk of demagnetization in the SPM machine is significantly higher. The demagnetization risk can be calculated using (5), where D is the demagnetization risk in percentage, H_{max} is the maximum field intensity, and H_c is the coercivity.

$$D = \frac{H_{max}}{H_c} \times 100 \quad (5)$$

Figs. 16 shows the demagnetization risk in percentage for different SPM and IPM motors with power ratings of 15 kW, 50 kW, 75 kW, 90 kW, 115 kW, and 125 kW in charging mode at two different temperatures. These power ratings are chosen due to their usage in commercial EVs [35]. Generally, it is observed that the demagnetization risk under IOBC increases with the increase in the power rating of the motor. At 100 °C, the risk of the IPM motor for all six power ratings is below 50 % making it extremely safe during charging the battery at rated power. On the other hand, the risk in SPM motors touches 80 % for 50 kW and over 95 % for 115- and 125-kW motors at 100 °C making it extremely vulnerable. Thus, charging at full power rating may not be possible in all conditions for such motors around 100 °C. Beyond 100 °C, the SPM machine will be severely irreversible demagnetized. In addition, extra care will be required to operate the motor within safe limits. Also, mitigation of demagnetization under unusual situations such as



(a)



(b)

Fig. 15. Magnetic flux density variation at the selected points on the PM charging mode. (a) SPM, (b) IPM

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overloading may not be guaranteed due to the thin margin. For example, Fig. 17 shows the comparison of the demagnetization risk at different charging levels for the 15 kW SPM and IPM machines. It has been noted that raising the charging level increases risk, however, this increase is not substantial. The overall risk for SPM motor is approximately 45% higher compared to the IPM. Therefore, it can be concluded that the SPM motor is vulnerable to demagnetization and thus this study adopted and manufactured only IPM motor for experimental study.

V. EXPERIMENTAL RESULTS

A. Experimental Setup

The simulation results show the superiority of the IPM motor over SPM in both propulsion and charging modes. Since the SPM motor shows more vulnerability toward irreversible demagnetization, only the IPM motor was manufactured for the targeted application. Thus, only the experimental results of the IPM motor will be discussed in this section. The entire experimental setup and the IPM motor are shown in Fig. 18(a) and (b), respectively. For the battery and grid, an emulator manufactured by Cinergia power solution S.L. is used [36]. The emulators are bidirectional AC and DC power supplies that have the capability to operate in constant current, constant voltage, impedance, and power modes. Besides, it can be used for battery testing and battery emulation. The control system was built in Simulink and dSPACE MicroLab box was used to conduct a hardware-in-loop experiment. A 40 kW six-phase inverter was used to drive the motor in propulsion and charging modes. An advanced 20 kW dynamo system equipped with National Instrument's myRIO along with other measurement and interface boards was used for data acquisition. To regulate and provide constant current for charging the battery and supplying power to the grid, a predictive current control

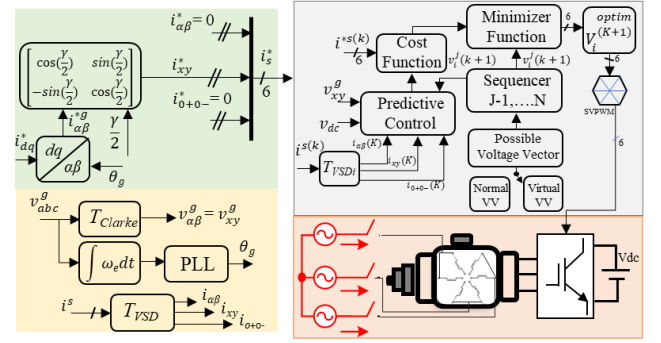


Fig. 19. Block diagram of the predictive current control algorithm during charging mode.

algorithm is employed [37]. Six-phase field-oriented control (FOC) algorithm is used to implement propulsion mode [38]. The charging mode was implemented using the algorithm shown in Fig. 19. The grid's current, i_d^* , is maximized while the quadrature component, i_q^* , is equated to zero to obtain the unity power factor.

By comparing the reference of the grid current components, the reference sequence current component is generated. The dq-axes current component is then transformed into an $\alpha\beta$ -reference frame using the angle obtained from the phase-lock-loop (PLL). In this way, synchronization with the grid is achieved. The reference values of $\alpha\beta$ and $0+0-$ components ($i_{\alpha\beta}^*$ and i_{0+0-}^*) of the currents are set to zero to eliminate the torque-producing component while the i_{xy}^* components are maximized to maximize the charging current. The regulation of the current components is performed using a predictive control, where a cost function is tuned to select optimal voltage vectors and optimal coefficients for suppressing and maximizing the current components. The dynamic response of the predictive control is outstanding compared to conventional proportional-resonant (PR) controllers.

B. Propulsion Mode Results

Experiments were conducted at rated speed and load to evaluate the performance of the benchmark IPMSM in propulsion mode. The electromagnetic torque and output power of the IPM motor in propulsion mode at the rated speed are shown in Fig. 20. When the motor was running at its rated speed of 1500 rpm, the load was applied in steps until the current reached its rated value. The torque value at the rated current is 103.4 Nm, which is slightly less than the FEM result. This is due to the limited frequency response of the torque sensor used in the experiment. This explains why the torque ripple cannot be detected either. Fig. 21(a) depicts the corresponding six-phase stator currents drawn by the motor to meet the load requirement. Fig. 21(b) depicts the motor's dynamic response during the step speed test. It is evident that the controller in use is appropriately reacting to the sudden change in speed. Overall, the IPM motor's performance in propulsion mode matches the simulation results.

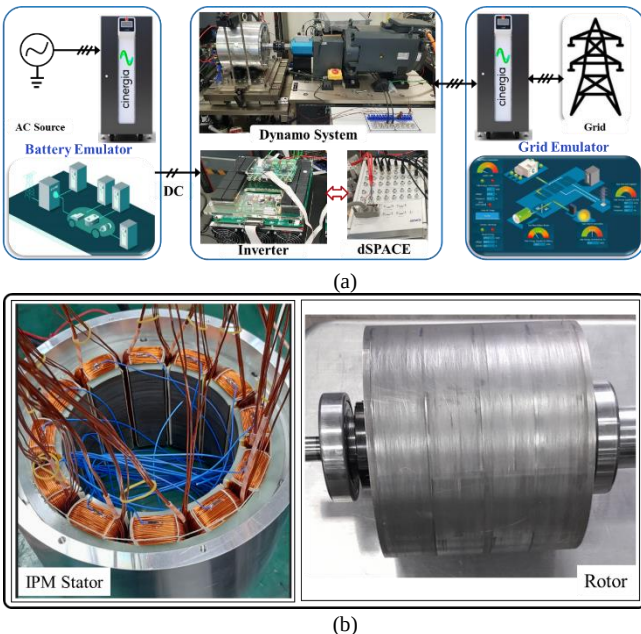


Fig. 18. Overall experimental Setup. (a) Motor dynamo, inverter, data acquisition system, grid, and battery emulators, (b) IPM motor.

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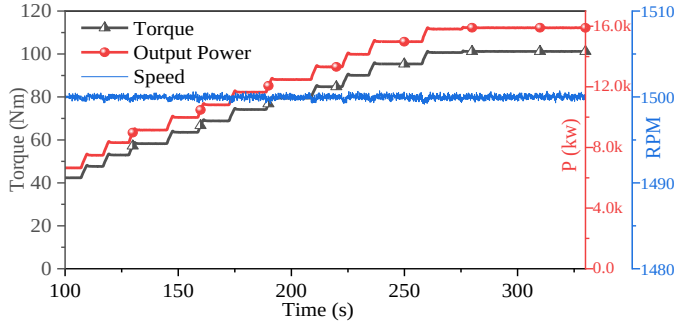


Fig. 20. Torque, output power, and speed of the IPM motor in propulsion mode.

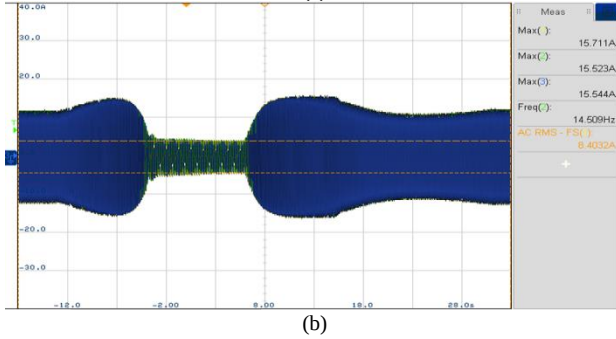
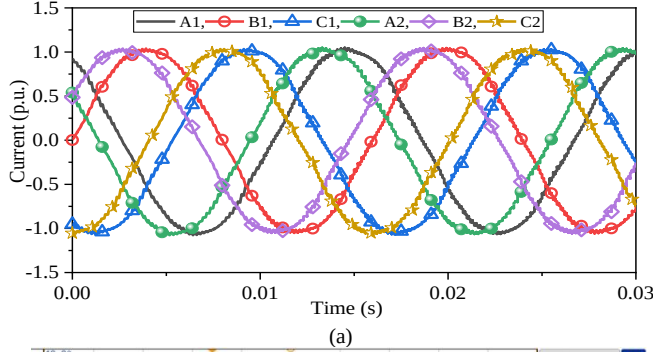


Fig. 21. Stator current at full load and rated speed under propulsion mode. (a) six phase currents, (b) stator current during the step speed test.

C. IOBC Mode Results

In the charging mode, the winding connection was changed according to Fig. 1(a) and connected to a three-phase grid. The stator currents of i_s^{a1} , i_s^{b2} , and the equivalent grid current, i_g^a in V2G mode for the IPM motor are shown in Fig. 22. The phase difference between i_s^{a1} and i_s^{b2} is 30° and with the grid current is 15° which is properly in line with the phasor diagram given in Fig. 1(c). Thus, the magnetic flux produced by each pair of winding cancelled out each other, and as a result, zero-average torque (no rotation) was achieved during charging mode. The grid current reached almost 1.93 times the phase current which is consistent with the method used in this study. Fig. 23 shows that the IPM motor has essentially no torque-producing $\alpha\beta$ current components, while the xy current components are balanced with noteworthy distortion. The phase and grid line currents are distorted for two reasons: the high ripple current component caused by the small value of the secondary subspace inductance in the case of asymmetrical winding, and the low order harmonic content caused by the nonzero-sequence voltage components of the $\alpha\beta$ and $0+0-$ subspaces. This results in a THD in the phase and line currents of around 12.3%. The

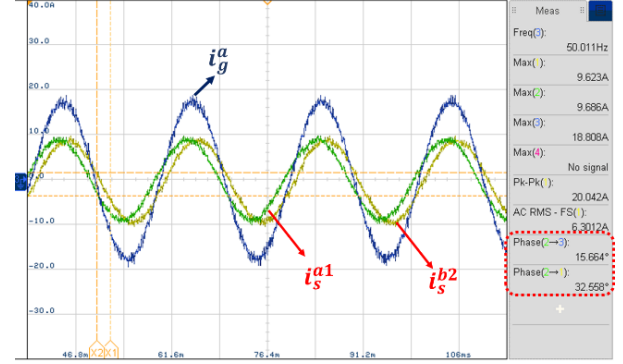


Fig. 22. Stator and grid current under charging mode.

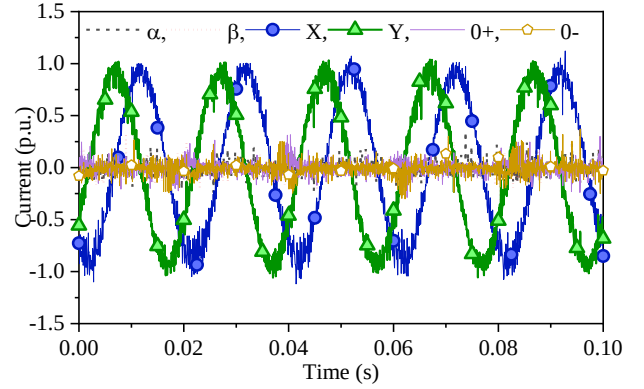


Fig. 23. $\alpha\beta$, xy , and $0+0-$ components of the stator current under charging.

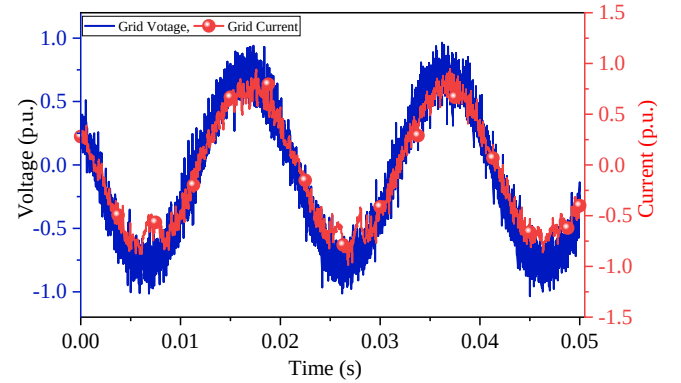


Fig. 24. Grid Phase voltage and current under charging mode.

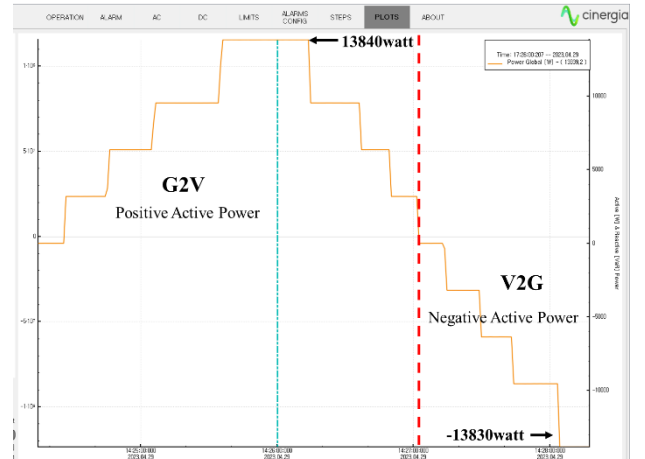


Fig. 25. Three-phase and net active power under V2G and G2V.

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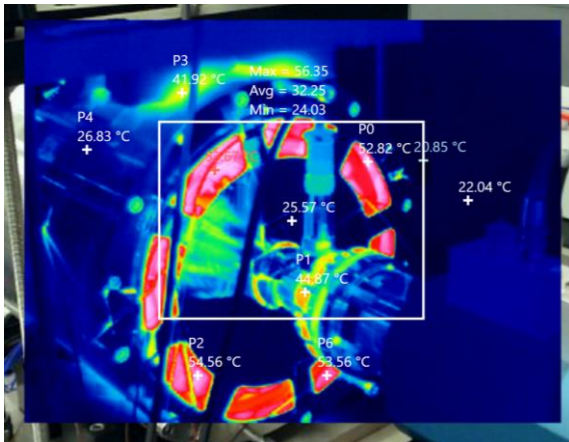
unity power factor is also ensured by keeping the phase voltage and current of the grid in phase, as shown in Fig. 24. Lastly, Fig. 25 depicts the three-phase and the net active power circulation between the battery and the grid through V2G and G2V. In the case of V2G, a negative active power indicates that the battery is sending power to the grid, whilst a positive active power indicates that the grid is supplying power to the battery.

D. Thermal Results

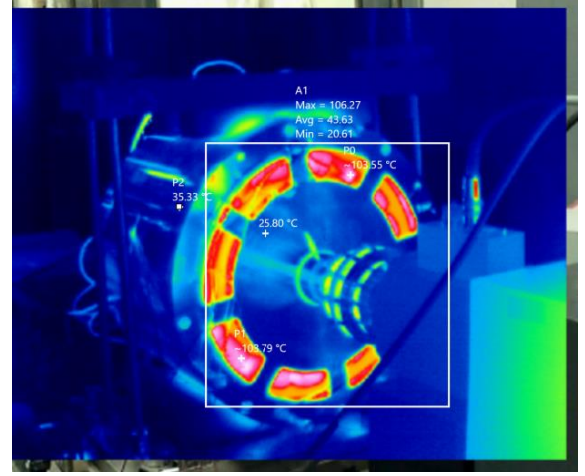
The thermal results of the 15 kW IPM motor were validated by operating the motor at full load in propulsion and charging at rated current. The temperature of the motor was monitored every five minutes until the rate of change of the temperature was close to zero. The rate of change of the temperature was significantly decreased after two and a half hours of operation indicating the steady-state region. The temperature was captured using an infrared camera. The temperature distribution during charging mode is given in Fig. 26(a), and during propulsion mode is displayed in Fig. 26(b). After two and half hours of charging, the winding temperature reached 56.3 °C and the housing temperature reached almost 38 °C. Similarly, in propulsion mode, the winding temperature reached almost 106 °C after operating at full load for two and a half hours. It is worth mentioning here that the rise time of the temperature curve in the propulsion mode was almost half compared to the charging mode. The experimental result of the rotor temperature is captured at the rotor's outer cross-section, which may differ from the temperature in the air gap in the center of the rotor. TABLE VI summarizes the experimental results under IOBC and propulsion mode. The experimental and simulation findings for winding temperatures are similar enough to demonstrate the correctness of the simulation results. As a result, we can depend on simulation results for PM

TABLE VI
EXPERIMENTAL RESULTS OF TEMPERATURE DISTRIBUTION IN IPM MOTOR

Ref. Temperature	Housing [°C]	Winding [°C]	Rotor lamination [°C]	Magnet [°C]
IOBC-Mode				
22°C	38	56.3	~28.8	-
Propulsion Mode				
22°C	72	106	~61.6	-



(a)



(b)

Fig. 26. IR camera and thermocouple captured temperature distribution in IPM-type machine during charging and propulsion mode. (a) Charging, (b) Propulsion.

temperature, which cannot be tested directly by experiment. Furthermore, we can also rely on the simulation results for the SPM motor.

E. Demagnetization Results

The health of the PMs in a PMSM cannot be measured directly, especially in the IPM-type machine where the PMs are buried in the rotor. The best way to diagnose the demagnetization is to analyze the input stator voltage and/or currents of the motor. To determine if any irreversible damage occurred in PM as a result of the machines being tested in charging mode for several hours, the frequency spectrum of the filtered input voltage was examined. It has been proven that, for FSCW PMSMs, the partial demagnetization results in additional even-order (2nd and 4th) harmonics in the input voltage as well as in the back-emf. Furthermore, when the severity of demagnetization increases, the magnitude of the odd-order harmonics (3rd and 5th) that are present in the healthy condition decreases [39].

The frequency spectrum of the input voltage for the SPM (simulation) and IPM motors are shown in Fig. 27. Results from the simulation and experiment are compared for the most dominant harmonics. Additionally, simulation outcomes for both machines subjected to irreversible demagnetization (single pole 50% demagnetized) are also displayed to demonstrate the behavior of both motors under PM demagnetization. In the presence of PM demagnetization, there are obvious reductions in the magnitude of the fundamental and odd-order harmonics, as well as additional 2nd and 4th-order harmonics appear. The comparison of experimental and simulation results in Fig. 27(a) demonstrates that the frequency spectrum of the input voltage is unchanged after operating the motors in charging mode for several hours. There is no notable difference in the voltage's fundamental component or any additional harmonics. This implies that the PMs are unaffected while in charging mode. Furthermore, the same severity of demagnetization has a higher detrimental effect on the SPM motor. The drop in the fundamental component of the input voltage is substantially greater in Fig. 27(b) compared to the IPM motor, illustrating the

susceptibility of the SPM motor. This experiment was performed at room temperature (22 °C), therefore, the worst-case scenario cannot be verified. The overall analysis suggests that there is no immediate threat of demagnetization as long as it operates within the safe limit of temperature. However, depending on the drive cycle, the PMs of the SPM machine may deteriorate faster compared to the IPM motor.

F. Discussion

PM demagnetization always remains a constraint in all types of PMSMs. IOBC adds extra risk to FSCW motors, thus, this study suggested adding the impact of IOBC in the optimization of the FSCW motors. An acceptable trade-off during optimization was adopted to maintain the propulsion mode performance and prevent PM demagnetization in the integrated charging mode. The findings show that the SPM motor has a higher risk of PM demagnetization in both propulsion and charging modes compared to the IPM motor. Apart from the temperature, the impact of the armature reaction in IOBC mode is more severe on the magnetic flux density of PMs in SPM motors, as shown in Fig. 12(b). The magnetic flux density significantly reduces in charging mode. On the other hand, the IPM motor is much safer. This allows the IPM motors to be used in charging mode at full rated power without any severe risk while the SPM motor may not be able to perform charging at full rated power. It is also found that the rate of change of the

temperature and the maximum value of the temperature is almost half in IOBC mode due to the absence of mechanical and air friction losses, thus, allowing the system to be used for longer periods for charging big-size battery packs. To ensure an acceptable range of efficiency, proper current control has a crucial role in minimizing the losses and, more importantly, the current ripple, because a higher current ripple negatively impacts the life of the battery. Issues related to galvanic isolation and safety could be tackled in future research.

VI. CONCLUSION

This article analyzed the impact of the inevitable sub, super, and slot harmonics on the PMs of FSCW-based PMSMs during the integrated charging process of EVs. A detailed PM demagnetization and thermal analysis of optimally designed SPM and IPM machines have been introduced under both propulsion and charging EV modes of operation. The performance of the PM machines in the IOBC process is the focus of this study, which has not been conceived so far. The asymmetrical six-phase 12-slot/10-pole winding configuration is adopted in this study.

First, the two machines are designed based on the efficient parametric MEC model and optimized using the multi-objective genetic algorithm considering IOBC. Six-phase IPM and SPM machines have been extensively compared from thermal and demagnetization perspectives. The former offers a higher average torque production in the propulsion at a lower PM volume. Whereas the latter exhibits lower core losses under both operational modes. Moreover, the PM loss is higher in the SPM machine than in the IPM one since the magnet volume is reduced when the IPM machine is employed. This is further proved by the temperature distributions of both machines, at which the IPM machine offers lower temperatures when compared to its SPM counterpart. Additionally, the rate-of-change of the heat and its maximum value in charging mode is almost half of the propulsion mode due to the absence of mechanical and air friction losses. Demagnetization risk increases as motor power rating and charging power rise. However, the risk in the SPM motor is over 45% higher than the risk in the IPM motor, allowing the IPM motor to be utilized in IOBC at full power. Experimental results have been carried out using a 15 kW IPM machine to validate the theoretical findings. Eventually, the G2V and V2G charging modes have been tested.

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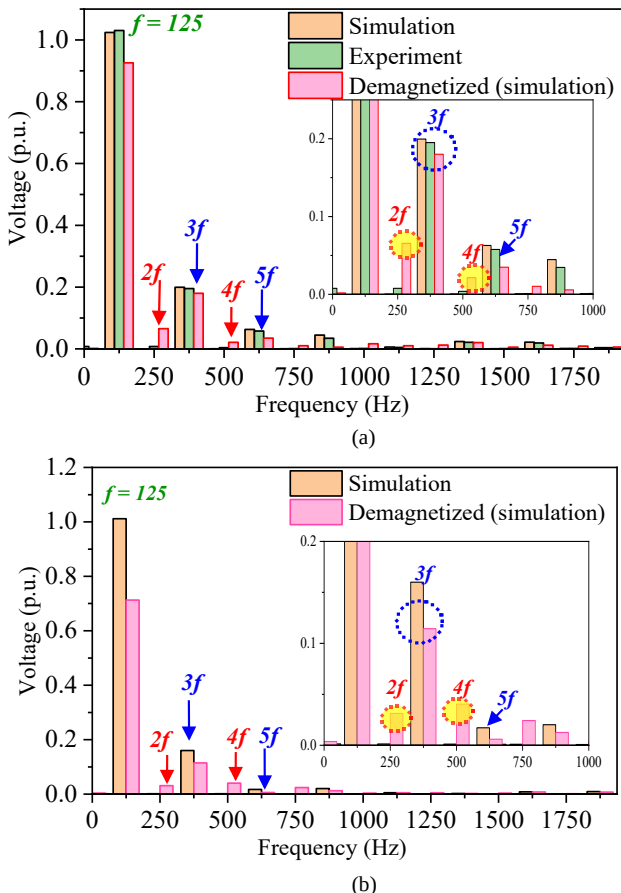


Fig. 27. Simulation and experimental results of the input voltage's frequency spectrums under healthy and demagnetization. (a) IPM machine. (b) SPM machine.

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